

FINAL PRACTICE!

Problem Solving with Computers-II

Link to class handout:

<https://bit.ly/one-problem-to-rule-them-all>

The image shows the C++ logo in a large, blue, 3D-style font. Below the logo is a snippet of C++ code in a monospaced font, with some words highlighted in color. The code is:

```
#include <iostream>
using namespace std;

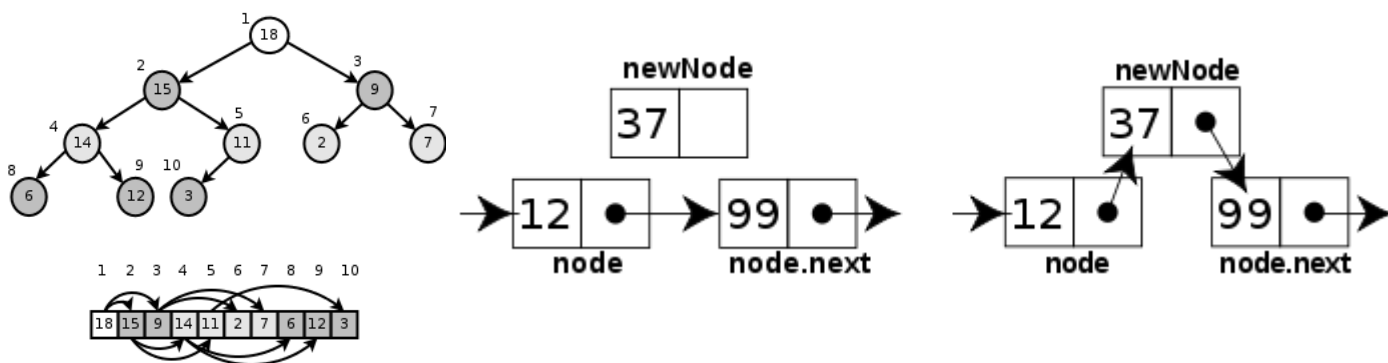
int main() {
    cout<<"Hola Facebook\n";
    return 0;
}
```

I can deal with pressure, and deadlines.



Resources for the Final Exam

- Leet code problems categorized to help you focus on the most relevant ones.
 - Those marked as “Optional challenge” won’t be on the final
- Review session led by LAs and TAs, this Sunday (over Zoom) at 10a
- Code from lectures: <https://github.com/ucsb-cs24-s25/cs24-s25-lectures>
- Past Exams: Available on Canvas
- Tool to visualize data structures: <https://visualgo.net/>



```

INSERTION-SORT(A)
1  for j = 2 to A.length
2    key = A[j]
3    // Insert A[j] into the sorted
   sequence A[1..j - 1].
4    i = j - 1
5    while i > 0 and A[i] > key
6      A[i + 1] = A[i]
7      i = i - 1
8    A[i + 1] = key
  
```

cost	times
c_1	n
c_2	$n - 1$
0	$n - 1$
c_4	$n - 1$
c_5	$\sum_{j=2}^n t_j$
c_6	$\sum_{j=2}^n (t_j - 1)$
c_7	$\sum_{j=2}^n (t_j - 1)$
c_8	$n - 1$

Data Structures and C++

Complexity Analysis

One problem to rule them all!

Problem (LP04): Longest Consecutive Sequence

Description: Given an unsorted array of integers `nums`, return the length of the longest consecutive elements sequence.

Try solving it in as many different ways as you can, then pick the best!

Ex: `nums = [100, 4, 200, 1, 3, 2]` `nums = [1, 0, 1, 2]`

Approach 1

Problem (LP04): Longest Consecutive Sequence

```
nums = [100, 4, 200, 1, 3, 2]
```

```
nums = [1, 0, 1, 2]
```

Approach 2

Problem (LP04): Longest Consecutive Sequence

```
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```

```
nums = [1, 0, 1, 2]
```

Approach 3

Problem (LP04): Longest Consecutive Sequence

```
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```

```
nums = [1, 0, 1, 2]
```

Approach 4 ?

Problem (LP04): Longest Consecutive Sequence

`nums = [100, 4, 200, 1, 3, 2]`

`nums = [1, 0, 1, 2]`

Reflection + Q&A

- Which approach did you try first, and why?
 - Did exploring other methods change how you thought about the problem?
- Which tool or idea resonated most with you?
 - If you had to solve this under time pressure, which approach would you pick and why?

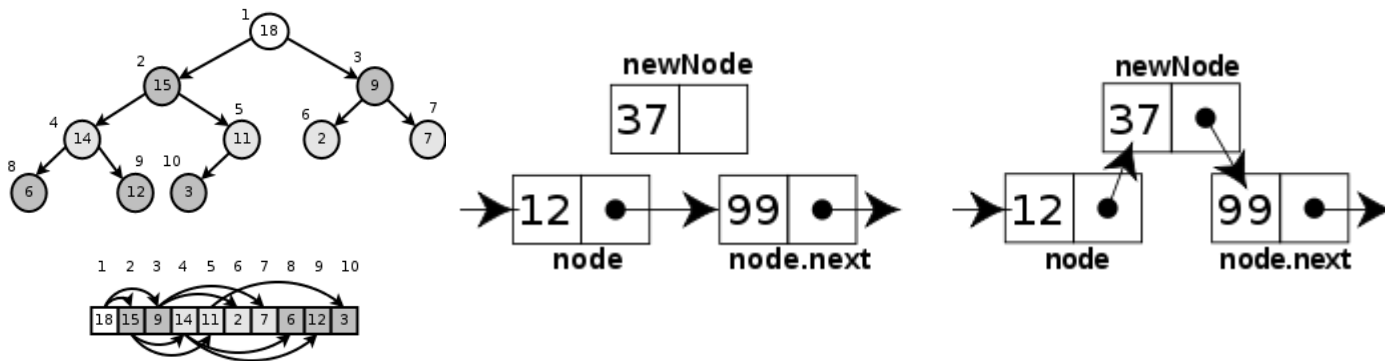
Questions?

CS24: Learning goals

Correct coding, clear thinking, no AI shortcuts

- Design and implement **larger programs** that **run fast**
- Organize **data** in programs using **data structures**
- **Analyze** the **complexity** of your programs
- Prep for **technical interviews**

Grow confident in your problem solving skills!



```

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Data Structures and C++

Complexity Analysis

Break: Please take a moment to fill the course evaluations!



PROBLEM SOLVING II

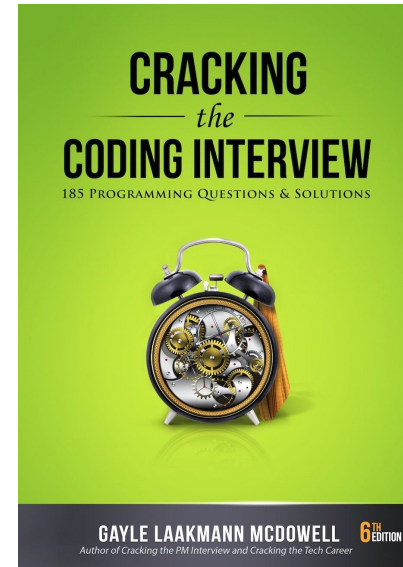
Student-FO

<https://go.blueja.io/WD3FDfJ2QUqKjETiskpCNw>

To access the evaluation, scan this QR code with your mobile phone.

Tips for Technical Interviews

1. Listen carefully
2. Draw an example
3. State the brute force or a partially correct solution
 - then work to get at a better solution
4. Optimize:
 - Make time-space tradeoffs to optimize runtime
 - Precompute information: Reorganize the data e.g. by sorting
5. Solidify your understanding of your algo before diving into writing code.
6. Start coding!



Thank you and all the best !

